Advanced Programming Project

Scenario

You are a Python Developer working in a cybersecurity company. A client requests a Python-based system that demonstrates different cryptographic techniques.

The system should be designed using Object-Oriented Programming concepts to ensure:

- Readability
- Maintainability
- Reusability
- Extensibility

Task.1

1. Explain the concept of **Inheritance** and its features.
2. Differentiate between types of inheritance:
    - Single
    - Multiple
    - Multilevel
    - Hierarchical
3. Explain **Polymorphism** in detail:
    - Method Overriding (Runtime Polymorphism in Python)
    - Method Overloading (Explain conceptually in Python)

Task.2

Part A: Block Cipher Modes

- Implement a **base abstract class** representing a block cipher.
- Create derived classes:
    - ECB
    - CBC
    - CTR

Each class must:

- Implement encryption and decryption methods
- Show different behavior depending on the mode

Part B: Key Generator

- Implement a class using **Linear Congruential Generator (LCG)** for key generation.
- The generated key should be used in:
  - Block cipher modes
  - RSA algorithm

Part C: RSA Algorithm

- Implement a simplified version of RSA:
  - Key generation
  - Encryption
  - Decryption
- Use the LCG class to assist in generating values.

Part D: SHA-1 Algorithm

- Implement SHA-1 hashing:
  - Takes input data
  - Produces a fixed-length hash
  - One-way function (no decryption)

Part E: Simple User Interface

- Design a simple console-based menu that allows the user to choose between:
  - ECB Mode
  - CBC Mode

- CTR Mode
- RSA Encryption
- SHA-1 Hash
- Exit

Task.3

1. Explain **Exception Handling**:
    - try / except / finally
    - Importance in secure systems
2. Define **UML (Unified Modeling Language)**
3. Explain types of UML diagrams:
    - Structural diagrams
    - Behavioral diagrams
4. Define **Class Diagram** and its components:
    - Classes
    - Attributes
    - Methods
    - Relationships

Task.4

1. Implement exception handling in your program to handle:
    - Invalid input
    - Encryption errors
    - Key generation issues
2. Analyze one real-world problem (choose one):
    - Login system
    - Library system
    - E-commerce system
3. Design a **Class Diagram** that includes:
    - Inheritance
    - Relationships between classes

Bonus (Optional)

- Compare between different algorithms in terms of performance
- Add more secure or advanced implementations
- Improve the user interface

This project aims to develop a mini encryption system that allows users to select among multiple cryptographic algorithms, implemented using object-oriented programming principles.